

Quiz 1

1 Grading Scheme

Question 1: Each question is for 1 point. Correct answers with incorrect or no justification will be given 0.5 points.

Question 2: 1 point if table is correct and 0.5 points if at least one part of the table is correct.

Question 3: 1 point if the table is correct and 0.5 points if there is at most one error.

Total grade is out of 7.

2 Solutions

Question 1:

1. False. $3 \times c = 100 \implies c = 100/3$, but $100/3$ is not an integer.
2. True. Let $c = -1$, then $-3 \times -1 = 3$.
3. False. For every integer c , $0 \times c = 0 \neq 4$.
4. True. Let $c = 0$, then $4 \times 0 = 0$.
5. True. For every integer c , $0 \times c = 0$ (Any integer satisfies the condition).

Question 2:

A	B	$A \implies B$	$\neg B$	$\neg A$	$\neg B \implies \neg A$
$\top$	$\top$	$\top$	$\perp$	$\perp$	$\top$
$\top$	$\perp$	$\perp$	$\top$	$\perp$	$\perp$
$\perp$	$\top$	$\top$	$\perp$	$\top$	$\top$
$\perp$	$\perp$	$\top$	$\top$	$\top$	$\top$

The two statements are equivalent.

Question 3:

$$(A \iff B) = ((A \implies B) \wedge (B \implies A))$$

A	B	$A \implies B$	$B \implies A$	$A \iff B$
$\top$	$\top$	$\top$	$\top$	$\top$
$\top$	$\perp$	$\perp$	$\top$	$\perp$
$\perp$	$\top$	$\top$	$\perp$	$\perp$
$\perp$	$\perp$	$\top$	$\top$	$\top$

3 Common Errors

- Answered Question 1 without justifications.
- Switched $a \mid b$ and $b \mid a$.
- Switched $\neg A$ and $\neg B$ in Question 2.
- Tried to show $(A \implies B) \iff (\neg A \implies \neg B)$.

A	B	$A \implies B$	$\neg A$	$\neg B$	$\neg A \implies \neg B$
$\top$	$\top$	$\top$	$\perp$	$\perp$	$\top$
$\top$	$\perp$	$\perp$	$\perp$	$\top$	$\top$
$\perp$	$\top$	$\top$	$\top$	$\perp$	$\perp$
$\perp$	$\perp$	$\top$	$\top$	$\top$	$\top$

These two statements are not equivalent, $(A \implies B) \not\iff (\neg A \implies \neg B)$.